



**Further Vocational Qualification in Information and
Communication Technology**

Sergei Ivanov

Ålands yrkesgymnasium

QUALIFICATION CERTIFICATE

Further Vocational Qualification in Information and
Communication Technology, 150 cp

Sergei Ivanov
020383-363U

Qualification Units

Scope, Competence Points Grade

Compulsory Units

Working in an operational environment for information and
communication technologies

30 Pass

Optional Units

Software design and software development projects

40 Pass

Software testing

40 Pass

Working as a software developer

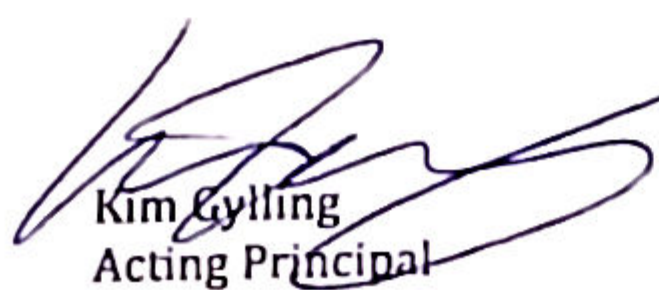
45 Pass

The scope of the qualification units for which the student has earned 155 competence points

Additional information:

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Mariehamn 23.08.2024


Kim Gylling
Acting Principal




Erika Eriksson
Vice Principal

The province of Åland has legislative competence for education in accordance with the Act on the Autonomy of Åland (1991:71), which is also the national law in Finland (FFS 1144/1991). The Åland Vocational Upper Secondary School belongs to Ålands gymnasium, which is an authority under the Government of Åland. The education to which this certificate refers is organised in accordance with The Provincial Act (2011:13) and decree (2011:75) on Upper Secondary Education (see also decree (2019/41) about change of upper secondary school), and The Finnish Act on Vocational Education and Training (531/2017).

The school's contact details:

Address	Ålands gymnasium PB 150 Ax-22101 Mariehamn Åland
Telephone	+358 18 536 200
E-mail	info@gymnasium.ax
Website	www.gymnasium.ax

The Finnish National Agency for Education's regulations of qualifications requirements Further Vocational Qualification in Information and Communication Technology (09.09.2020, OPH-2639-2020) has been followed in the programme. The qualification is a decree approved by the Ministry of Education and Culture on the structure of qualifications in vocational education and training (680/2017) and is taken after basic education.

Grading scale/Pass requirement:

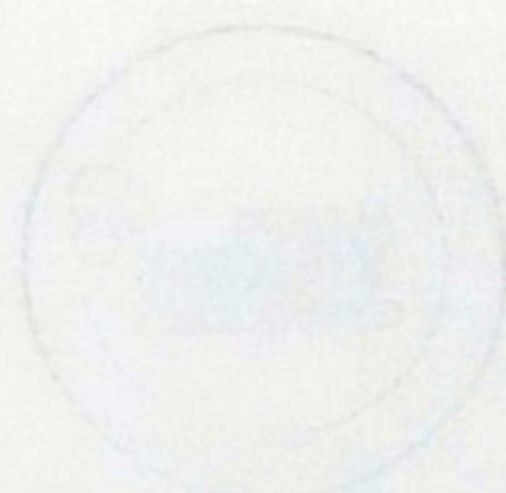
Pass/Fail

Definition of the scope of the qualification and competence points

The scope of a further vocational qualification is 120, 150 or 180 competence points and the scope of a specialist vocational qualification is 160, 180 or 210 competence points.

Access to the next level of education/training

The qualification confers eligibility for higher education studies.





CERTIFICATE SUPPLEMENT



Title of the qualification (FI, SV)
<i>Tieto- ja viestintätekniikan ammattitutkinto</i> <i>Yrkesexamen i informations- och kommunikationsteknik</i> <i>Qualification requirements entered into force on 01.01.2021 (OPH-2639-2020)</i>

Translated title of the qualification (EN)
<i>Further vocational qualification in Information and Communication Technology</i>
This translation has no legal status.

Profile of skills and competences
Composition of the qualification The scope of the further vocational qualification in information and communication technology is 150 competence points. The qualification is composed of a unit that is compulsory for all candidates (30 competence points) and optional units (120 competence points). One or more units (40 competence points) from another vocational upper secondary qualification or further or specialist vocational qualification can be included in the qualification. Vocational skills and competences required for completion of the qualification Holders of a further vocational qualification in information and communication technology have the vocational skills required to independently plan and implement an assignment according to the customer's wishes in work tasks and working environments in the information and communication technology sector. They work adeptly, promoting cooperation with different stakeholders. They follow the rapidly changing sector, update their knowledge and develop their competence. They network and share information with other professionals of their field.

Range of occupations accessible to the holder of the certificate	
Holders of a further vocational qualification in information and communication technology can carry out diverse work tasks in the information and communication technology sector. Depending on the studies they have chosen, qualification holders can specialise in specialist and consultation tasks concerning data networks, data systems, cyber security, welfare technology, electronics production, data analytics and machine learning or software development.	
Official basis of the certificate	
Name and status of the body awarding the certificate An education provider with an authorisation to provide education and award qualifications granted by the Ministry of Education and Culture.	Name and status of the national/regional authority providing accreditation/recognition of the certificate Ministry of Education and Culture Finnish National Agency for Education, the national agency subordinate to the Ministry of Education and Culture
Level of the certificate (national or international) National Qualifications Framework (nqf) 4 European Qualifications Framework (eqf) 4 ISCED 3	Grading scale/Pass requirements pass/fail
Access to the next level of education/training The qualification confers eligibility for higher education studies.	International agreements -
Legal basis Vocational Education and Training Act 531/2017 and the decrees that complement it.	

Officially recognised ways of obtaining the certificate
The vocational skills required for completing the modules are demonstrated by performing practical tasks in authentic situations and work processes (demonstration of knowledge and skills). The student's knowledge and skills are assessed diversely by a teacher and a working life representative and compared to the competence defined in the qualification requirements. The education provider awards the student a qualification certificate once the qualification has been completed acceptably in accordance to the qualification requirements laid down by the Finnish National Agency for Education.
Entry requirements No formal qualifications are required to take the competence test.
Additional information Finnish National Agency for Education P.O. Box 380, FI-00531 Helsinki, Finland www.oph.fi

Explanatory note

This document is designed to provide additional information about the specified certificate and does not have any legal status in itself. The format of the description is based on the following texts: Council Resolution 93/C 49/01 of 3 December 1992 on the transparency of qualifications, Council Resolution 96/C 224/04 of 15 July 1996 on the transparency of vocational training certificates, and Recommendation 2001/613/EC of the European Parliament and of the Council of 10 July 2001 on mobility within the Community for students, persons undergoing training, volunteers, teachers and trainers.

For more information on transparency, visit <http://www.cedefop.europa.eu/en/themes/understanding-qualifications>

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ÅLAND ISLANDS

CERTIFICATE OF COMPLETION

Sergei Ivanov

date of birth | 02.03.1983

has completed the
2-year grit:lab program on the Åland Islands 2022-2024

with a **specialization in**
UX/UI

The program is based on the 01Edu peer-to-peer learning curriculum

Mariehamn, Åland Islands, 23.08.2024



Anna-Lena Svenblad
Director, grit:lab



Kim Gylling
Principal, Ålands yrkesgymnasium

1. Go Module

No.	Completed mandatory activities	Experience points (k.B)	Description
1.1	Go Piscine	-	Introduction to the fundamentals of Golang
1.2	Go Reloaded	5.00	Build a text processor
1.3	ASCII Art	6.13	Build a text-to-ASCII graphic converter
1.4	ASCII Art Web	9.20	Host a web GUI for the text-to-ASCII graphic converter
1.5	Groupe Tracker	24.5	Build a website to visualize artist performance date
1.6	Lem In	34.4	Build a digital version of an ant farm
1.7	Forum	76.3	Build a web forum

2. JavaScript Module

No.	Completed mandatory activities	Experience points (k.B)	Description
2.1	Javascript Piscine	-	Introduction to the fundamentals of JavaScript
2.2	Make Your Game	147.00	Create a 60-FPS single-player game
2.3	Real-Time Forum	104.00	Create a single-page application forum
2.4	GraphQL	100.00	Create a hosted profile page using GraphQL to query data
2.5	Social Network	480.00	Create a Facebook-like social network

Appendix: List of completed mandatory modules
Sergei Ivanov
Date of birth: 02.03.1983

No.	Completed mandatory activities	Experience points (k.B)	Description
2.6	Mini Framework	145.00	Create a custom framework that abstracts the DOM
2.7	Bomberman DOM	319.00	Create a Bomberman-like multiplayer game

3. Rust Module

No.	Completed mandatory activities	Experience points (k.B)	Description
3.1	Rust piscine	-	Introduction to the Fundamentals of Rust
3.2	Smart Road	75.3	Create a new traffic control algorithm for autonomous vehicles
3.3	Filler	105.0	Create a player to win an algorithmic game against the predefined robot
3.4	RT	672.0	Create a ray tracer to render 3D scenes into 2D images
3.5	Local host	386.0	Create a reliable HTTP/1.1 server
3.6	Multiplayer FPS	1680.0	Create a client-server multiplayer version of the game Maze Wars
3.7	0 Shell	696.0	Create a simple shell

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Appendix: List of completed mandatory modules
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4. UX/UI Module

No.	Completed mandatory activities	Experience points (k.B)	Description
4.1	Piscine UX	140.003	Introduction to the Fundamentals of User Experience
4.2	Piscine UI	105.0	Introduction to the Fundamentals of User Interfaces
4.3	A Table	201.25	Design a food delivery app
4.4	Get A Room	70.0	Design a room booking service on desktop
4.5	Let's Do Some Sports	75.25	Design a sport app with data tracking on mobile and e-watch
4.6	Let's Fair Trade	161.0	Design an e-commerce platform for second-hand clothes on desktop and mobile

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